Discussion Section Worksheet 1 - Solutions

PSTAT 5A, Fall 2026

1. What is your TA's name and when are their office hours?

Solution: Ziheng Wei. Office hour: Thursday, 6:30–8:00 PM on Zoom.

2. Is each of the following values a legitimate probability value? Explain any "no" answers.

a) -0.56

Solution: No. A probability must be between 0 and 1, inclusive.

b) 1.05

Solution: No. A probability cannot be greater than 1.

c) 0.95

Solution: Yes.

d) 0.77

Solution: Yes.

e) 0.00

Solution: Yes. A probability of 0 corresponds to an impossible event.

3. A fair coin is flipped three times.

a) Define the sample space, S, for this experiment.

Solution: $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. There are $2^3 = 8$ equally likely outcomes.

b) What is the probability that you observe no tails?

Solution: No tails means HHH, so $P(\text{no tails}) = \frac{1}{8}$.

c) What is the probability that you observe three heads or three tails?

Solution: The favorable outcomes are HHH and TTT, so

$$P(\text{three heads or three tails}) = \frac{2}{8} = \frac{1}{4}.$$

4. You are given a choice of whipped cream, syrup, or fruit for your pancakes. Which one would you pick? Your TA will survey your entire section to help you answer the following questions.

a) How many students are in your section?

Solution: 21 students.

b) How many are female?

Solution: 13 students.

c) How many chose whipped cream?

Solution: 1 student.

d) How many chose syrup?

Solution: 6 students.

5. a) Using the information collected above, fill in the frequency counts in the following table.

	Female	**Female^C**	**Total**
Whipped cream	1	0	1
Syrup	3	3	6
Fruit	9	5	14
Total	13	8	21

5. b) Now, fill in the corresponding probabilities in the following table.

	Female	**Female^C**	**Total**
Whipped cream	$\frac{1}{21}$	0	$\frac{1}{21}$
Syrup	$\frac{3}{21} = \frac{1}{7}$	$\frac{3}{21} = \frac{1}{7}$	$\frac{6}{21} = \frac{2}{7}$
Fruit	$\frac{9}{21} = \frac{3}{7}$	$\frac{5}{21}$	$\frac{14}{21} = \frac{2}{3}$
Total	$\frac{13}{21}$	$\frac{8}{21}$	1

Solution: Each probability is the corresponding frequency count divided by the total number of students, 21.

6. Suppose a student in your section is chosen at random.

For notation, let W = chose whipped cream, S = chose syrup, F = chose fruit, and G = female.

a) What is the probability that they chose a topping other than whipped cream or syrup?

Solution: "Other than whipped cream or syrup" means neither whipped cream nor syrup. Since the only three choices are whipped cream, syrup, and fruit, this is the event F. Equivalently, it is $(W \cup S)^C$.

$$P(F) = \frac{14}{21} = \frac{2}{3}.$$

b) What is the probability that they are female or chose fruit?

Solution: This is $P(G \cup F)$. Use the addition rule so that students who are both female and chose fruit are not counted twice:

$$\begin{aligned} P(G \cup F) &= P(G) + P(F) - P(G \cap F) \\ &= \frac{13}{21} + \frac{14}{21} - \frac{9}{21} \\ &= \frac{18}{21} = \frac{6}{7}. \end{aligned}$$

c) If the student chose syrup, what is the probability they are female?

Solution: This is the conditional probability $P(G \mid S)$. Once we know the student chose syrup, the relevant group consists of the 6 students who chose syrup. Of those, 3 are female:

$$P(G \mid S) = \frac{P(G \cap S)}{P(S)} = \frac{3/21}{6/21} = \frac{3}{6} = \frac{1}{2}.$$

d) What is the probability that they chose whipped cream and are not female?

Solution: This is $P(W \cap G^C)$. The frequency table shows that no student was both in Female^C and chose whipped cream:

$$P(W \cap G^C) = \frac{0}{21} = 0.$$

7. Chat with your classmates. Make new friends!

Solution: No numerical solution.